Smart Study Room Digital Twin: Spatial Monitoring and Decision Support

Afrin Jubaida

This prototype demonstrates how a photorealistic representation of an indoor study room can be combined with real-time simulated environmental monitoring and explainable decision support. The room was captured as a Gaussian-splat reconstruction, cleaned and compressed, and rendered interactively in a web browser using Babylon.js. Five spatially anchored overlays represent temperature, humidity, occupancy, illumination, and computer power consumption. Their values are realistically simulated with bounded random variation and updated continuously, while a transparent rule-based layer converts the readings into practical recommendations.

The visual foundation of the prototype is 3D Gaussian Splatting, which represents a scene using optimized anisotropic Gaussian primitives and enables high-quality novel-view rendering at interactive rates [1]. This approach is useful for digital-twin applications because a real indoor space can be reconstructed from imagery without manually modelling every wall, object, and surface. In the prototype, the Gaussian-splat reconstruction provides a navigable spatial interface through which environmental and operational conditions can be interpreted in their physical context.

The system is also informed by the broader digital-twin concept. Fuller et al. describe digital twins as virtual representations supported by enabling technologies such as Internet of Things sensing, data analytics, artificial intelligence, and communication infrastructure [2]. A meaningful digital twin therefore extends beyond a static 3D model by connecting the virtual representation to changing information about the physical environment. In the built-environment context, Sacks et al. explain that digital-twin information systems can integrate monitoring data with intelligent functions to provide accurate status information and support proactive analysis and operational optimization [5]. Although the present prototype uses simulated rather than physical sensor readings, its architecture follows these principles by integrating a spatial model, continuously changing data, and a decision-support process. It should therefore be understood as a digital-twin proof of concept rather than a fully synchronized operational twin.

The selected variables are relevant to indoor environmental quality and smart-building operation. Temperature and humidity influence thermal comfort, illumination affects the usability of a workspace, occupancy provides context for interpreting energy consumption, and device power reveals potential operational waste. Research on healthy indoor digital twins highlights the role of building information modelling, IoT sensing, analytics, and smart control in supporting healthier and more responsive indoor environments [3]. Building-focused digital-twin research has also demonstrated how environmental, occupancy, and equipment information can support intelligent building operation, energy management, and occupant comfort [4].

The prototype implements four controlled scenarios: normal conditions, low light, an empty room with active equipment, and a crowded room. Example rules recommend increasing desk lighting when an occupied workspace falls below 300 lux, turning off equipment when the room is empty, and increasing ventilation or cooling when temperature or humidity exceeds configured thresholds. The crowded-room scenario can produce several recommendations simultaneously. Because every recommendation contains both an action and an explanation, the decision logic remains interpretable and suitable for demonstration to non-technical facility users.

A practical business-to-business application would be a subscription-based smart-building monitoring service for universities, libraries, offices, co-working spaces, hotels, or property-management companies. A service provider could scan client spaces, create browser-accessible spatial twins, and connect them to real sensors and building-management systems. Facility managers could remotely inspect rooms, identify comfort or lighting problems, detect unnecessary energy consumption, and prioritize maintenance, ventilation, or HVAC interventions. The service could be priced per room, building, or monitored asset.

Future development would include physical sensor integration, historical data storage, multi-room dashboards, automated alerts, predictive comfort and energy models, and secure integration with lighting and HVAC controls. Privacy, cybersecurity, sensor reliability, data interoperability, and validation of decision thresholds would also need to be addressed before real deployment. Overall, the prototype demonstrates how photorealistic spatial computing, environmental monitoring, and transparent decision support can be integrated into a lightweight browser-based service.

References

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